

Algebraic Geometry 19

Riemann surfaces

Contents

1 Problem 15: The Riemann Surface Associated with $w^2 = z^3 - z$

Statement

Let

$$f(z) = z^3 - z,$$

and consider the subset

$$R = \{(z, w) \in \mathbb{C}^2 : w^2 = f(z)\}.$$

Let

$$\pi : R \rightarrow \mathbb{C}$$

be the restriction of the projection onto the first factor.

Show that R has the structure of a Riemann surface, on which π is analytic.

If g denotes the projection onto the second factor, show that g is also analytic.

By deleting three appropriate points from R , show that π yields a covering map from the resulting Riemann surface

$$R_0 \rightarrow \mathbb{C} \setminus \{-1, 0, 1\},$$

and that R_0 is analytically isomorphic to the Riemann surface associated with the complete analytic function

$$(z^3 - z)^{1/2}$$

over

$$\mathbb{C} \setminus \{-1, 0, 1\}.$$

Solution

Consider

$$F(z, w) = w^2 - z^3 + z.$$

Then

$$R = F^{-1}(0).$$

Step 1: Smoothness of R

Compute the partial derivatives:

$$F_z = -3z^2 + 1, \quad F_w = 2w.$$

A singular point would satisfy

$$F = 0, \quad F_z = 0, \quad F_w = 0.$$

From

$$F_w = 0$$

we obtain

$$w = 0.$$

Substituting into

$$F = 0$$

gives

$$z^3 - z = 0,$$

hence

$$z \in \{-1, 0, 1\}.$$

Evaluate F_z :

$$F_z(-1, 0) = -2, \quad F_z(0, 0) = 1, \quad F_z(1, 0) = -2.$$

Thus

$$F_z \neq 0$$

at every point where $F = F_w = 0$.

Therefore R has no singular points.

By the complex implicit function theorem,

$$R$$

is a one-dimensional complex manifold.

Hence

$$\boxed{R \text{ is a Riemann surface.}}$$

Step 2: Analyticity of the Projections

The maps

$$\pi(z, w) = z, \quad g(z, w) = w$$

are restrictions of the coordinate projections

$$\mathbb{C}^2 \rightarrow \mathbb{C}.$$

Since coordinate projections are holomorphic,

$$\boxed{\pi \text{ and } g \text{ are analytic maps on } R.}$$

Step 3: Branch Points

The equation

$$w^2 = z^3 - z = z(z - 1)(z + 1)$$

shows that the two values

$$w = \pm \sqrt{z^3 - z}$$

coincide precisely when

$$z^3 - z = 0.$$

Hence the branch points occur at

$$z = -1, \quad 0, \quad 1.$$

Correspondingly the points

$$(-1, 0), \quad (0, 0), \quad (1, 0)$$

are ramification points of the projection.

Define

$$R_0 = R \setminus \{(-1, 0), (0, 0), (1, 0)\}.$$

Step 4: Covering Map

Let

$$U = \mathbb{C} \setminus \{-1, 0, 1\}.$$

Since

$$z^3 - z$$

has no zeros on U , a local branch of

$$\sqrt{z^3 - z}$$

can be chosen on every sufficiently small simply connected neighborhood.

Therefore over such a neighborhood $V \subset U$,

$$\pi^{-1}(V) = V_+ \sqcup V_-,$$

where

$$V_{\pm} = \{(z, \pm\sqrt{z^3 - z}) : z \in V\}.$$

Each component projects biholomorphically onto V .

Thus

$$\boxed{\pi : R_0 \rightarrow \mathbb{C} \setminus \{-1, 0, 1\}}$$

is a two-sheeted covering map.

Step 5: Relation with the Complete Analytic Function

The multivalued function

$$(z^3 - z)^{1/2}$$

has two local branches

$$+\sqrt{z^3 - z}, \quad -\sqrt{z^3 - z}.$$

Analytic continuation around the branch points interchanges these branches.

The associated complete analytic function is obtained by gluing together all branches.

Its Riemann surface is precisely the surface obtained from

$$w^2 = z^3 - z.$$

Hence

R_0 is analytically isomorphic to the Riemann surface of $(z^3 - z)^{1/2}$.

2 Problem 16: Radius of Convergence and Singular Boundary Points

Statement

Let

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

be a power series of radius of convergence 1.

For $w \in D(0, 1)$, let

$$\rho(w)$$

denote the radius of convergence of the Taylor expansion of f about w .

Show that a point

$$\zeta, \quad |\zeta| = 1,$$

is regular if and only if

$$\rho(\zeta/2) > \frac{1}{2}.$$

Suppose furthermore that all coefficients a_n are nonnegative real numbers.

Show that

$$|f^{(r)}(\zeta/2)| \leq f^{(r)}(1/2)$$

for all r , and hence

$$\rho(\zeta/2) \geq \rho(1/2).$$

Deduce that 1 is a singular point.

Solution

Step 1: Characterization of Regular Boundary Points

Let

$$|\zeta| = 1.$$

Observe that

$$\left| \zeta - \frac{\zeta}{2} \right| = \frac{1}{2}.$$

Suppose

$$\rho(\zeta/2) > \frac{1}{2}.$$

Then the Taylor series centered at

$$\zeta/2$$

converges in a disc extending beyond the point ζ .

Hence f is analytic in a neighborhood of ζ .

Therefore ζ is regular.

Conversely, if ζ is regular, then f extends analytically across a neighborhood of ζ .

Combining this neighborhood with the unit disc gives a domain containing a disc centered at

$$\zeta/2$$

of radius strictly larger than

$$\frac{1}{2}.$$

Hence

$$\rho(\zeta/2) > \frac{1}{2}.$$

Thus

$\zeta \text{ is regular} \iff \rho(\zeta/2) > \frac{1}{2}.$
--

Step 2: Estimate for Derivatives

Assume

$$a_n \geq 0$$

for all n .

The r -th derivative is

$$f^{(r)}(z) = \sum_{n=r}^{\infty} n(n-1) \cdots (n-r+1) a_n z^{n-r}.$$

Evaluating at

$$z = \frac{\zeta}{2},$$

gives

$$f^{(r)}(\zeta/2) = \sum_{n=r}^{\infty} n(n-1) \cdots (n-r+1) a_n \left(\frac{\zeta}{2}\right)^{n-r}.$$

Taking absolute values and using

$$|\zeta| = 1,$$

we obtain

$$|f^{(r)}(\zeta/2)| \leq \sum_{n=r}^{\infty} n(n-1) \cdots (n-r+1) a_n \left(\frac{1}{2}\right)^{n-r}.$$

The right-hand side is precisely

$$f^{(r)}(1/2).$$

Hence

$$\boxed{|f^{(r)}(\zeta/2)| \leq f^{(r)}(1/2).}$$

Step 3: Comparison of Radii

The radius of convergence of the Taylor series at w satisfies

$$\frac{1}{\rho(w)} = \limsup_{r \rightarrow \infty} \left| \frac{f^{(r)}(w)}{r!} \right|^{1/r}.$$

Applying the previous estimate yields

$$\limsup_{r \rightarrow \infty} \left| \frac{f^{(r)}(\zeta/2)}{r!} \right|^{1/r} \leq \limsup_{r \rightarrow \infty} \left| \frac{f^{(r)}(1/2)}{r!} \right|^{1/r}.$$

Therefore

$$\frac{1}{\rho(\zeta/2)} \leq \frac{1}{\rho(1/2)},$$

and hence

$$\boxed{\rho(\zeta/2) \geq \rho(1/2)}.$$

Step 4: Deduction that 1 is Singular

Suppose, for contradiction, that 1 is regular.

Then by Step 1,

$$\rho(1/2) > \frac{1}{2}.$$

By Step 3,

$$\rho(\zeta/2) \geq \rho(1/2) > \frac{1}{2}$$

for every

$$|\zeta| = 1.$$

Applying Step 1 again, every point of the unit circle would be regular.

Hence f would extend analytically across the entire unit circle.

Since the unit circle is compact, f would then be analytic in a neighborhood of the closed unit disc.

This would imply that the original power series has radius of convergence strictly greater than 1, contradicting the hypothesis.

Therefore

$$\boxed{1 \text{ is a singular point of } f.}$$

This is a form of *Pringsheim's theorem*: for a power series with nonnegative coefficients and finite radius of convergence, the positive real boundary point is necessarily singular.

3 Theory Behind Problems 15–16: Algebraic Curves, Riemann Surfaces, Elliptic Curves, and Singularities of Power Series

Problems 15 and 16 represent a major transition in complex analysis.

Up to this point, most constructions involved analytic continuation, coverings, monodromy, and complex tori.

These two problems open two of the deepest branches of modern mathematics:

$$\boxed{\text{Algebraic Curves} \implies \text{Riemann Surfaces} \implies \text{Elliptic Curves}}$$

and

$$\boxed{\text{Power Series} \implies \text{Singularities} \implies \text{Analytic Continuation} \implies \text{Pringsheim's Theorem.}}$$

The first direction is geometric.

The second direction is analytic.

Together they form two central pillars of modern complex analysis.

3.1 1. Algebraic Curves as Riemann Surfaces

Problem 15 begins with the equation

$$w^2 = z^3 - z.$$

At first sight this appears to be merely an equation in two variables.

However it defines a geometric object:

$$R = \{(z, w) \in \mathbb{C}^2 : w^2 = z^3 - z\}.$$

Such sets are called algebraic curves.

More generally,

$$P(z, w) = 0$$

defines an algebraic curve whenever P is a polynomial.

Examples include

$$w = z^2,$$

$$w^2 = z,$$

$$w^2 = z^3 - z,$$

$$w^2 = z^5 - z.$$

These objects lie at the heart of algebraic geometry.

3.2 2. Why Multivalued Functions Appear

The equation

$$w^2 = z^3 - z$$

can be rewritten as

$$w = \pm \sqrt{z^3 - z}.$$

For a generic value of z , there are two possible values of w .

Hence the function is naturally two-valued.

The ordinary complex plane is therefore insufficient to make the function single-valued.

Riemann's revolutionary idea was:

Instead of changing the function, change the space.

Construct a new space on which the function becomes single-valued.

That new space is precisely the Riemann surface

R .

Thus

Riemann Surfaces Resolve Multivaluedness.

3.3 3. Branch Points

The two sheets merge whenever

$$z^3 - z = 0.$$

Thus

$$z = -1, \quad 0, \quad 1.$$

These are called branch points.

Near such a point z_0 ,

$$w^2 \approx z - z_0,$$

and therefore

$$w \approx \sqrt{z - z_0}.$$

Locally the geometry is identical to the familiar square-root surface.

The two sheets exchange when one travels around a branch point.

This phenomenon is called monodromy.

Branch Points $\implies$ Sheet Interchange $\implies$ Monodromy.
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3.4 4. Smooth Algebraic Curves

Consider an algebraic curve

$$P(z, w) = 0.$$

If

$$\nabla P = \left(\frac{\partial P}{\partial z}, \frac{\partial P}{\partial w} \right) \neq 0,$$

then the implicit function theorem applies.

Locally one may solve for w as a holomorphic function of z or vice versa.

Thus every smooth algebraic curve automatically becomes a Riemann surface.

This is one of the most important bridges in mathematics:

Smooth Algebraic Curves $\iff$ Compact Riemann Surfaces.
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The equivalence is one of the foundational theorems of modern geometry.

3.5 5. Coverings and Complete Analytic Functions

Removing the branch points gives

$$R_0 = R \setminus \{(-1, 0), (0, 0), (1, 0)\}.$$

The projection

$$\pi : R_0 \rightarrow \mathbb{C} \setminus \{-1, 0, 1\}$$

becomes a two-sheeted covering map.

This is exactly the covering associated with the complete analytic function

$$(z^3 - z)^{1/2}.$$

The two sheets correspond to

$$+\sqrt{z^3 - z}$$

and

$$-\sqrt{z^3 - z}.$$

Analytic continuation around branch points exchanges them.

Hence

Complete Analytic Functions $\iff$ Covering Riemann Surfaces.

3.6 6. Monodromy and Permutation of Sheets

Every loop in

$$\mathbb{C} \setminus \{-1, 0, 1\}$$

acts on the sheets of the covering.

For square roots the action is

$$(+) \leftrightarrow (-).$$

Thus the monodromy group is

$$\mathbb{Z}_2.$$

More generally, for equations such as

$$w^n = P(z),$$

one obtains larger permutation groups.

This eventually leads to:

$$\boxed{\text{Monodromy Theory} \implies \text{Galois Theory.}}$$

Indeed, modern Galois theory can be viewed as an algebraic analogue of monodromy.

3.7 7. Genus and Topology

The surface

$$w^2 = z^3 - z$$

is not merely a covering.

Its topology is highly nontrivial.

The Riemann–Hurwitz formula implies

$$g = 1.$$

Hence the surface has genus one.

Topologically it is a torus:

$$\boxed{w^2 = z^3 - z \implies g = 1 \implies \text{Torus.}}$$

This is the first appearance of an elliptic curve.

3.8 8. Elliptic Curves

An elliptic curve is a nonsingular curve of the form

$$y^2 = x^3 + ax + b.$$

Problem 15 gives the special case

$$y^2 = x^3 - x.$$

Elliptic curves appear in:

- Number theory,
- Fermat's Last Theorem,
- Modular forms,
- Cryptography,
- String theory,
- Arithmetic geometry.

Thus Problem 15 is actually an introduction to one of the most important objects in modern mathematics.

3.9 9. Radius of Convergence and Singularities

Problem 16 concerns the fundamental relationship between power series and singularities. Suppose

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n.$$

The central theorem states:

Theorem 3.1. *The radius of convergence equals the distance from z_0 to the nearest singularity of f .*

Thus singularities completely determine convergence.

3.10 10. Examples of Radius Determination

For

$$f(z) = \frac{1}{1 - z},$$

the nearest singularity to 0 is

$$z = 1.$$

Hence

$$R = 1.$$

For

$$f(z) = \frac{1}{1 + z^2},$$

the singularities are

$$z = \pm i.$$

Again

$$R = 1.$$

Thus convergence radii are geometric quantities.

Radius of Convergence = Distance to Nearest Singularity.
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3.11 11. The Local Radius Function

Problem 16 introduces

$$\rho(w).$$

This quantity measures the radius of convergence of the Taylor expansion centered at w . Geometrically,

$$\rho(w) = \text{distance from } w \text{ to the nearest singularity.}$$

Hence

$$\rho(w)$$

encodes geometric information about the singular set.

3.12 12. Analytic Continuation Through Boundary Points

Let

$$|\zeta| = 1.$$

The point

$$\frac{\zeta}{2}$$

lies exactly

$$\frac{1}{2}$$

units from ζ .

Therefore

$$\rho(\zeta/2) > \frac{1}{2}$$

means that the Taylor disc centered at

$$\zeta/2$$

extends beyond the unit circle.

Consequently f continues analytically through ζ .

Thus

$$\rho(\zeta/2) > \frac{1}{2} \iff \zeta \text{ is a regular boundary point.}$$

3.13 13. Positivity of Coefficients

The second half of Problem 16 assumes

$$a_n \geq 0.$$

This assumption is surprisingly powerful.

Indeed,

$$|a_n z^n| \leq a_n |z|^n.$$

Therefore

$$|f(z)| \leq f(|z|).$$

Positivity allows comparison of values throughout the disc using only the positive real axis.

This creates strong monotonicity properties unavailable for arbitrary power series.

3.14 14. Pringsheim's Theorem

The final conclusion of Problem 16 is a famous theorem.

Theorem 3.2 (Pringsheim). *If*

$$f(z) = \sum_{n=0}^{\infty} a_n z^n, \quad a_n \geq 0,$$

has finite radius of convergence R , then

$$z = R$$

is a singular point.

In the present problem,

$$R = 1.$$

Therefore

$$z = 1$$

must be singular.

3.15 15. Why Pringsheim's Theorem is Surprising

For a general power series this need not be true.

Consider

$$\sum_{n=0}^{\infty} (-1)^n z^n = \frac{1}{1+z}.$$

The radius of convergence is

$$1,$$

but the singularity lies at

$$-1,$$

not at 1.

Thus positivity of coefficients imposes a remarkable geometric restriction.

Positive Coefficients $\implies$ Positive Real Singular Boundary Point.

3.16 16. Connection with Analytic Combinatorics

Generating functions often have nonnegative coefficients:

$$F(z) = \sum_{n \geq 0} c_n z^n.$$

Examples include counting:

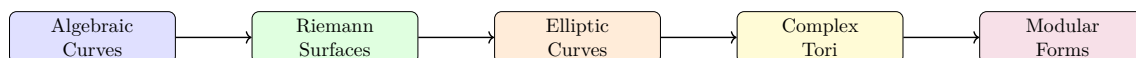
- Trees,
- Graphs,
- Permutations,
- Lattice paths,
- Tilings.

Pringsheim's theorem guarantees that the dominant singularity occurs on the positive real axis.

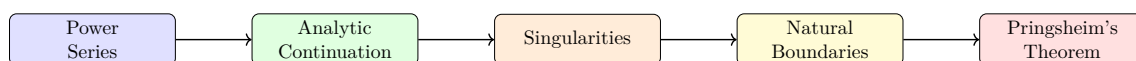
This fact is one of the foundations of analytic combinatorics.

3.17 Bigger Picture

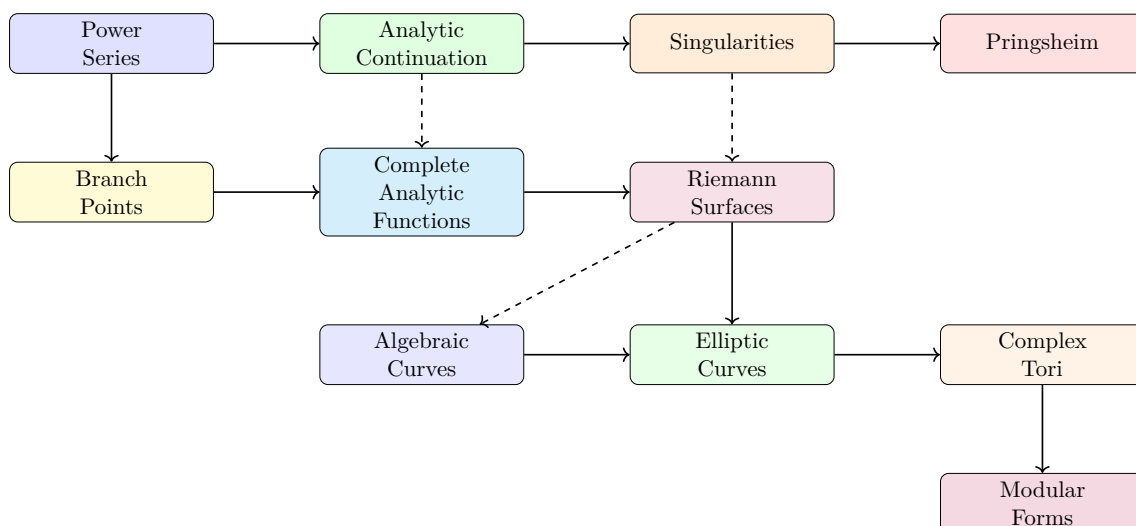
Problem 15 leads toward geometry:



Problem 16 leads toward analysis:



3.18 Grand Synthesis



Thus Problems 15–16 represent the point where the theory splits into the two great directions of modern complex analysis:

Geometry of Riemann Surfaces and Theory of Analytic Continuation
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4 Detailed Examples for Problems 15–16

The following examples illustrate the geometric and analytic ideas underlying Problems 15 and 16. They show how algebraic curves become Riemann surfaces and how singularities determine the behavior of power series.

4.1 Example 1: The Surface $w^2 = z^3 - z$

Consider

$$R = \{(z, w) \in \mathbb{C}^2 : w^2 = z^3 - z\}.$$

Since

$$z^3 - z = z(z - 1)(z + 1),$$

the special values of z are

$$z = -1, \quad z = 0, \quad z = 1.$$

For example, if

$$z = 2,$$

then

$$w^2 = 2^3 - 2 = 6.$$

Hence

$$w = \pm\sqrt{6}.$$

Thus over the point $z = 2$ there are two points on the surface:

$$(2, \sqrt{6}), \quad (2, -\sqrt{6}).$$

Therefore the surface consists of two sheets lying above the z -plane.

$$\boxed{\text{Two values of } w \iff \text{Two sheets of the surface.}}$$

4.2 Example 2: What Happens at a Branch Point?

Take

$$z = 1.$$

Then

$$w^2 = 1^3 - 1 = 0,$$

so

$$w = 0.$$

The two branches

$$+\sqrt{z^3 - z}, \quad -\sqrt{z^3 - z}$$

coincide.

To study the local geometry, write

$$z = 1 + \xi.$$

Then

$$z^3 - z = (1 + \xi)^3 - (1 + \xi) = 2\xi + 3\xi^2 + \xi^3.$$

Thus

$$w^2 = 2\xi + O(\xi^2).$$

Consequently

$$w = \sqrt{2} \sqrt{\xi} + O(\xi^{3/2}).$$

Equivalently,

$$w \approx \sqrt{2} \sqrt{z - 1}.$$

This is exactly the local model of a square-root branch point.

After one circuit around $z = 1$,

$$\sqrt{z-1} \mapsto -\sqrt{z-1}.$$

Hence

$$w \mapsto -w.$$

Therefore one loop exchanges the two sheets.

4.3 Example 3: Local Coordinates Near a Ramification Point

Consider the point

$$(1, 0) \in R.$$

Near this point the coordinate z is not suitable because two sheets merge.

Instead use

$$t = w.$$

Define

$$F(z, w) = w^2 - z^3 + z.$$

Then

$$F_z = -3z^2 + 1.$$

At

$$(1, 0),$$

we have

$$F_z = -2 \neq 0.$$

Therefore the implicit function theorem implies that

$$z$$

can be written as a holomorphic function of

$$w.$$

Expanding near $w = 0$,

$$z = 1 + \frac{1}{2}w^2 + \cdots.$$

Hence

$$z = 1 + \frac{1}{2}t^2 + \cdots.$$

The projection

$$\pi(z, w) = z$$

becomes

$$t \mapsto 1 + \frac{1}{2}t^2 + \cdots.$$

Thus locally the covering looks like

$$t \mapsto t^2,$$

which is the standard ramified covering.

4.4 Example 4: Removing Branch Points Produces a Covering

Define

$$R_0 = R \setminus \{(-1, 0), (0, 0), (1, 0)\}.$$

Let

$$U = \mathbb{C} \setminus \{-1, 0, 1\}.$$

Choose a small simply connected domain

$$V \subset U.$$

Since

$$z^3 - z$$

has no zeros on V , there exists a holomorphic branch

$$h(z) = \sqrt{z^3 - z}.$$

Then

$$\pi^{-1}(V) = V_+ \sqcup V_-,$$

where

$$V_+ = \{(z, h(z)) : z \in V\},$$

and

$$V_- = \{(z, -h(z)) : z \in V\}.$$

Each sheet projects biholomorphically onto V .

Therefore

$$\boxed{\pi : R_0 \rightarrow \mathbb{C} \setminus \{-1, 0, 1\}}$$

is a two-sheeted covering map.

4.5 Example 5: Monodromy Around Branch Points

Choose the branch

$$w = \sqrt{z^3 - z}.$$

A loop around any one of

$$-1, \quad 0, \quad 1$$

changes

$$w \mapsto -w.$$

Thus the action on sheets is

$$(+) \leftrightarrow (-).$$

Hence the monodromy group is

$$\mathbb{Z}_2.$$

The covering transformation is

$$(z, w) \longmapsto (z, -w).$$

This is the deck transformation exchanging the two sheets.

4.6 Example 6: Computing the Genus

Compactify the curve by adding the point at infinity.

The resulting map

$$\overline{R} \rightarrow \widehat{\mathbb{C}}$$

has degree 2 and four branch points:

$$-1, \quad 0, \quad 1, \quad \infty.$$

Apply the Riemann–Hurwitz formula:

$$2g(\overline{R}) - 2 = 2(2g(\widehat{\mathbb{C}}) - 2) + \sum (e_p - 1).$$

Since

$$g(\widehat{\mathbb{C}}) = 0,$$

and each branch point contributes 1,

$$2g(\overline{R}) - 2 = 2(-2) + 4.$$

Thus

$$2g(\overline{R}) - 2 = 0.$$

Therefore

$$g(\overline{R}) = 1.$$

Hence

$$\boxed{w^2 = z^3 - z}$$

defines a torus.

It is therefore an elliptic curve.

4.7 Example 7: Radius of Convergence from Singularities

Consider

$$f(z) = \frac{1}{1-z}.$$

Its Taylor expansion at 0 is

$$f(z) = \sum_{n=0}^{\infty} z^n.$$

The nearest singularity is

$$z = 1.$$

Since

$$|1 - 0| = 1,$$

the radius of convergence is

$$R = 1.$$

Now expand about

$$w = \frac{1}{2}.$$

The nearest singularity is still 1, so

$$\rho(1/2) = \left|1 - \frac{1}{2}\right| = \frac{1}{2}.$$

Thus

$$\boxed{\rho(1/2) = \frac{1}{2}.$$

4.8 Example 8: A Regular Boundary Point

Consider

$$f(z) = \frac{1}{1+z}.$$

Its Taylor series at 0 is

$$\sum_{n=0}^{\infty} (-1)^n z^n.$$

The radius of convergence is

$$1,$$

because the singularity occurs at

$$z = -1.$$

Take

$$\zeta = 1.$$

Then

$$\zeta/2 = \frac{1}{2}.$$

The nearest singularity to $1/2$ is -1 .

Hence

$$\rho(1/2) = \left| \frac{1}{2} - (-1) \right| = \frac{3}{2}.$$

Since

$$\frac{3}{2} > \frac{1}{2},$$

the point 1 is regular.

Thus

$$\boxed{\rho(\zeta/2) > \frac{1}{2} \implies \zeta \text{ is regular.}}$$

4.9 Example 9: Pringsheim's Theorem

Consider

$$f(z) = \sum_{n=0}^{\infty} z^n = \frac{1}{1-z}.$$

All coefficients satisfy

$$a_n = 1 \geq 0.$$

The radius of convergence is

$$R = 1.$$

Pringsheim's theorem states that

$$z = R$$

must be singular.

Indeed,

$$z = 1$$

is a pole.

Thus

$$\boxed{a_n \geq 0 \implies z = R \text{ is singular.}}$$

4.10 Example 10: Another Positive-Coefficient Example

Consider

$$f(z) = \sum_{n=0}^{\infty} 2^n z^n.$$

Summing the geometric series,

$$f(z) = \frac{1}{1 - 2z}.$$

The radius of convergence is

$$R = \frac{1}{2}.$$

The singularity occurs exactly at

$$z = \frac{1}{2}.$$

Again the positive real boundary point is singular.

This is another illustration of Pringsheim's theorem.

4.11 Example 11: Catalan Numbers

Let

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

The generating function is

$$F(z) = \sum_{n=0}^{\infty} C_n z^n = \frac{1 - \sqrt{1 - 4z}}{2z}.$$

The singularity occurs when

$$1 - 4z = 0.$$

Hence

$$z = \frac{1}{4}.$$

The radius of convergence is

$$R = \frac{1}{4}.$$

The singularity is not a pole but a branch point.

Thus Pringsheim's theorem applies even to non-pole singularities.

4.12 Example 12: Why Positivity Matters

Consider

$$f(z) = \sum_{n=0}^{\infty} (-1)^n z^n = \frac{1}{1+z}.$$

The radius of convergence is

$$1.$$

However

$$z = 1$$

is perfectly regular:

$$f(1) = \frac{1}{2}.$$

The singularity occurs at

$$z = -1.$$

Thus without positivity the conclusion of Pringsheim's theorem fails.

Therefore the condition

$$a_n \geq 0$$

is essential.

4.13 Final Interpretation

Problem 15 teaches:

Multivalued algebraic functions become single-valued on Riemann surfaces.

Problem 16 teaches:

The radius of convergence is controlled by singularities,

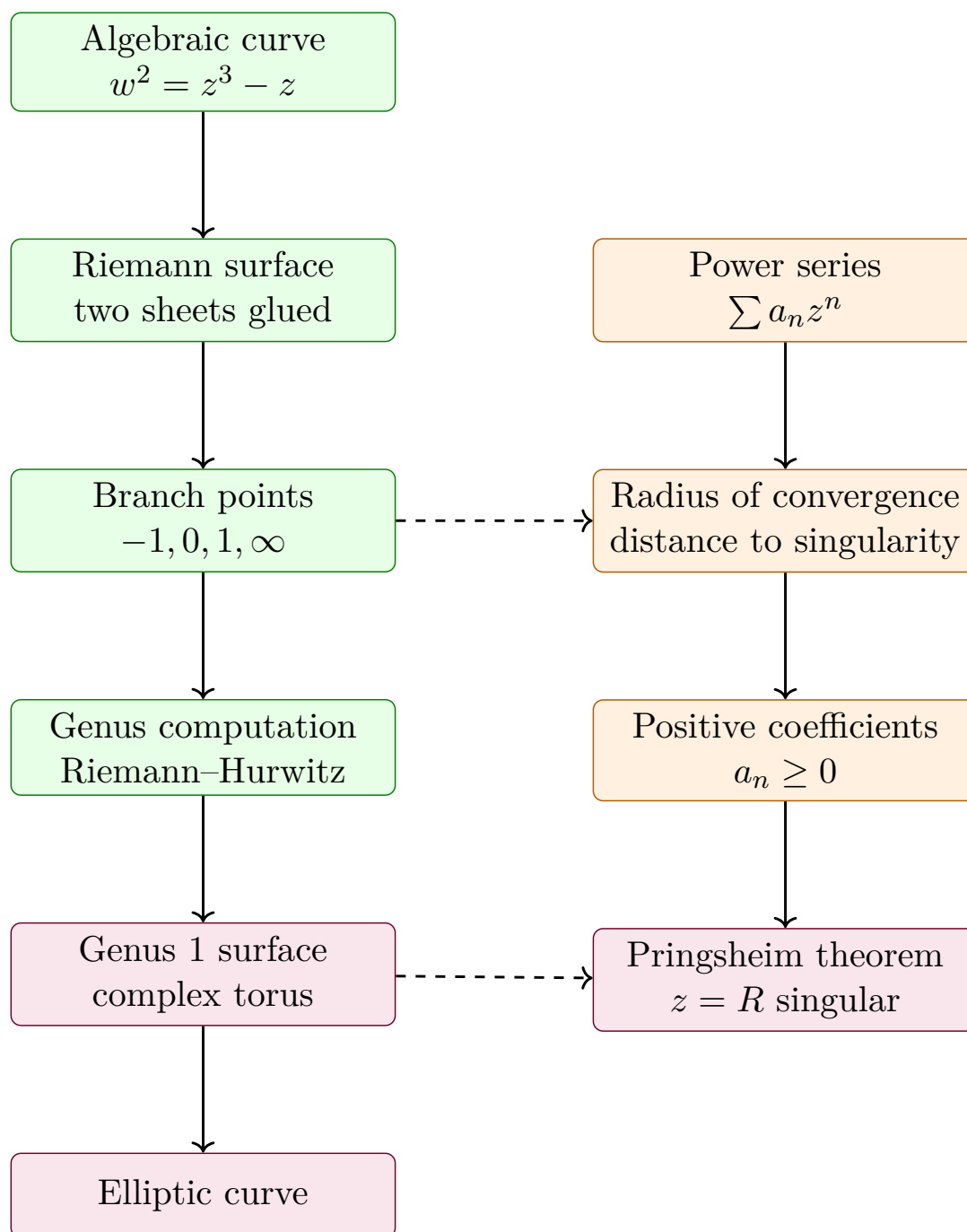
and

Positive coefficients force a singularity at the positive real boundary point.

Together these problems connect the geometry of algebraic curves with the analytic theory of power series and singularities.

5 Colored Diagrams, Tables, and Visual Schemes for Problems 15–16

5.1 Geometric Roadmap



5.2 Branch Points of $w^2 = z^3 - z$

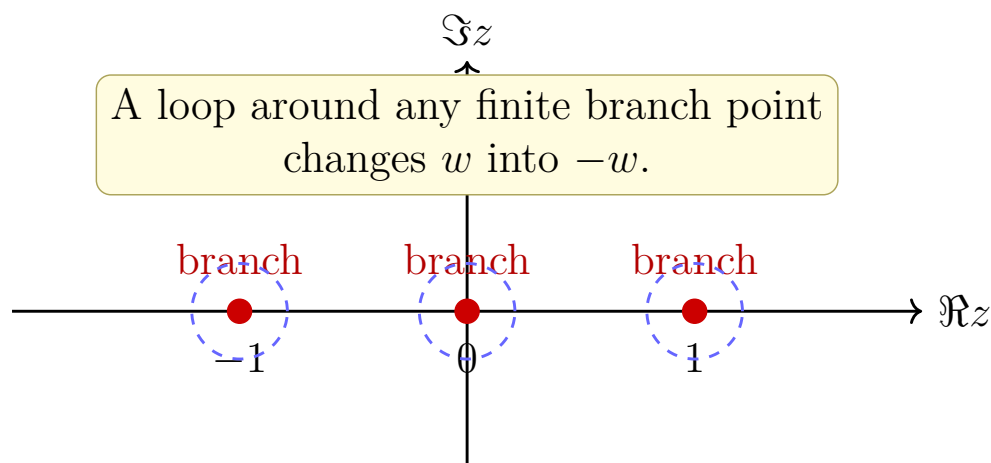
Since

$$z^3 - z = z(z - 1)(z + 1),$$

the finite branch points are

$$-1, \quad 0, \quad 1.$$

The compactified curve also has a branch point at infinity.

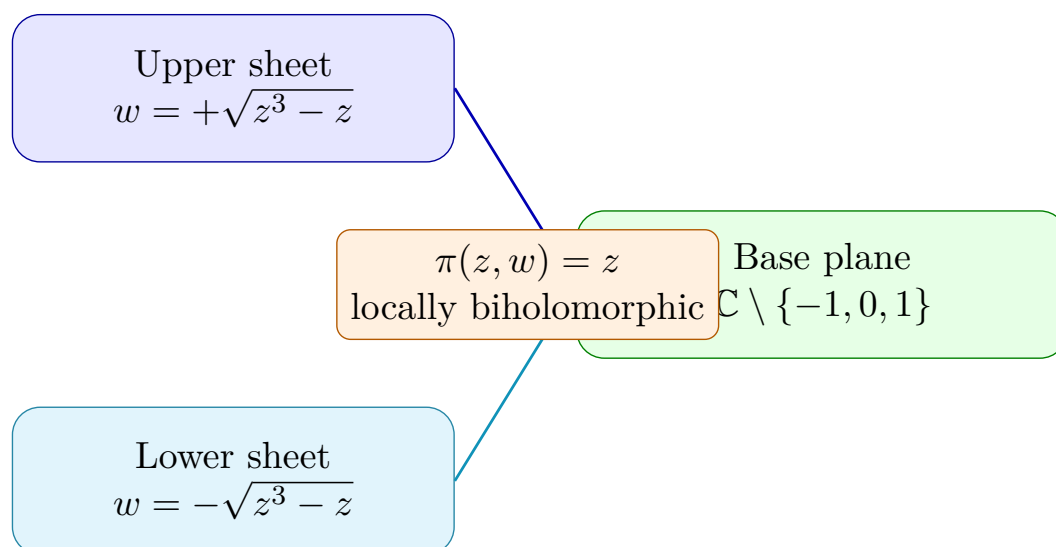


5.3 Two-Sheeted Covering Picture

Away from the branch points, the projection

$$\pi : R_0 \rightarrow \mathbb{C} \setminus \{-1, 0, 1\}$$

is a two-sheeted covering.



5.4 Local Ramification Model

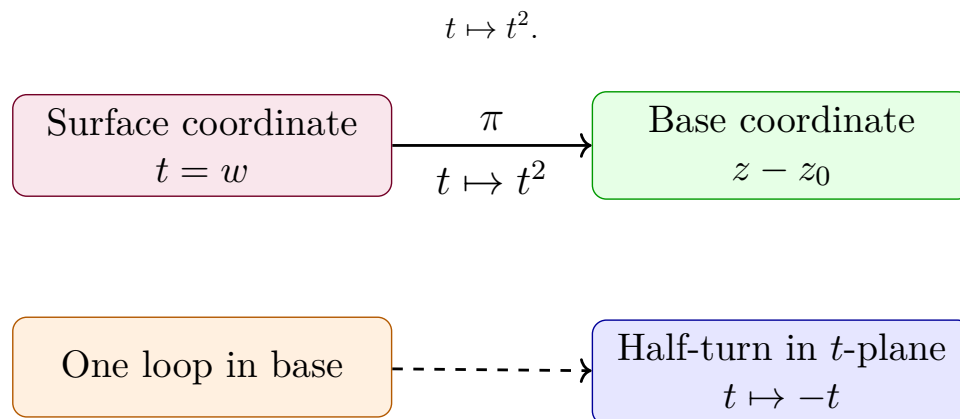
Near a branch point z_0 , the curve has the local form

$$w^2 \sim z - z_0.$$

Using $t = w$ as a local coordinate,

$$z = z_0 + ct^2 + \cdots.$$

Thus locally the projection behaves like



5.5 Riemann–Hurwitz Genus Calculation

For the compactified curve

$$\overline{R}: \quad w^2 = z^3 - z,$$

the map

$$\overline{R} \rightarrow \widehat{\mathbb{C}}$$

has degree 2 and four branch points:

$$-1, \quad 0, \quad 1, \quad \infty.$$

Therefore

$$2g(\overline{R}) - 2 = 2(2g(\widehat{\mathbb{C}}) - 2) + 4.$$

Since

$$g(\widehat{\mathbb{C}}) = 0,$$

we get

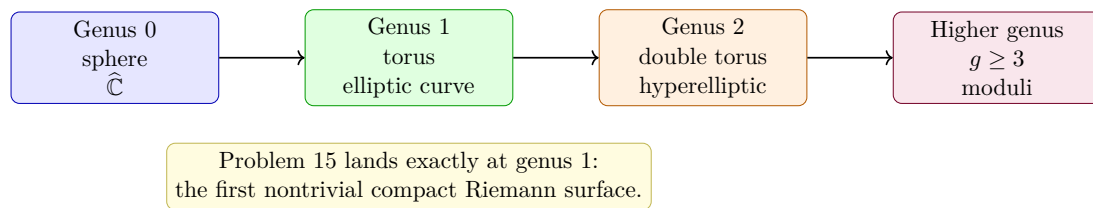
$$2g(\overline{R}) - 2 = 2(-2) + 4 = 0.$$

Hence

$$g(\overline{R}) = 1.$$

Object	Degree	Branch points	Genus
$\widehat{\mathbb{C}}$	1	none	0
$w^2 = z$	2	$0, \infty$	0
$w^2 = z^3 - z$	2	$-1, 0, 1, \infty$	1
$w^2 = z^5 - z$	2	six branch points	2

5.6 Genus Ladder



5.7 Radius of Convergence Geometry

For a power series centered at 0, the radius of convergence is the distance to the nearest singularity.

For a Taylor expansion centered at w , the radius is denoted

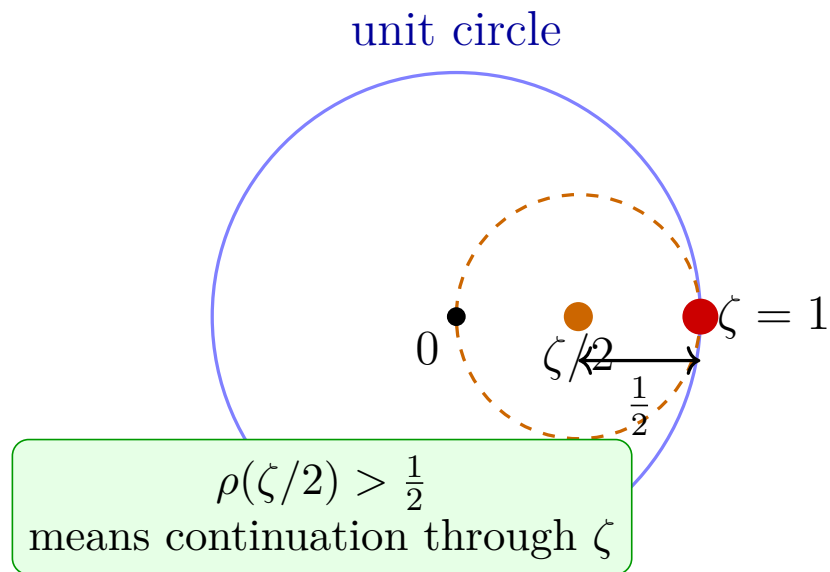
$$\rho(w).$$

In Problem 16, the key point is

$$w = \frac{\zeta}{2}, \quad |\zeta| = 1.$$

Then

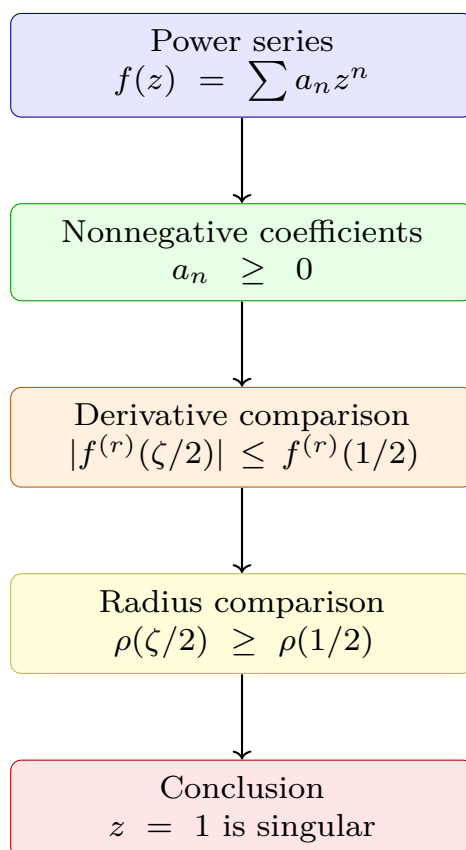
$$\left| \zeta - \frac{\zeta}{2} \right| = \frac{1}{2}.$$



5.8 Regular Versus Singular Boundary Point

Condition	Geometric meaning	Boundary behavior
$\rho(\zeta/2) > \frac{1}{2}$	Taylor disc crosses the unit circle	ζ is regular
$\rho(\zeta/2) = \frac{1}{2}$	Taylor disc touches a singular obstruction	ζ is singular
$\rho(\zeta/2) < \frac{1}{2}$	Singularity occurs before reaching ζ	ζ cannot be reached from that center

5.9 Pringsheim's Theorem Flowchart



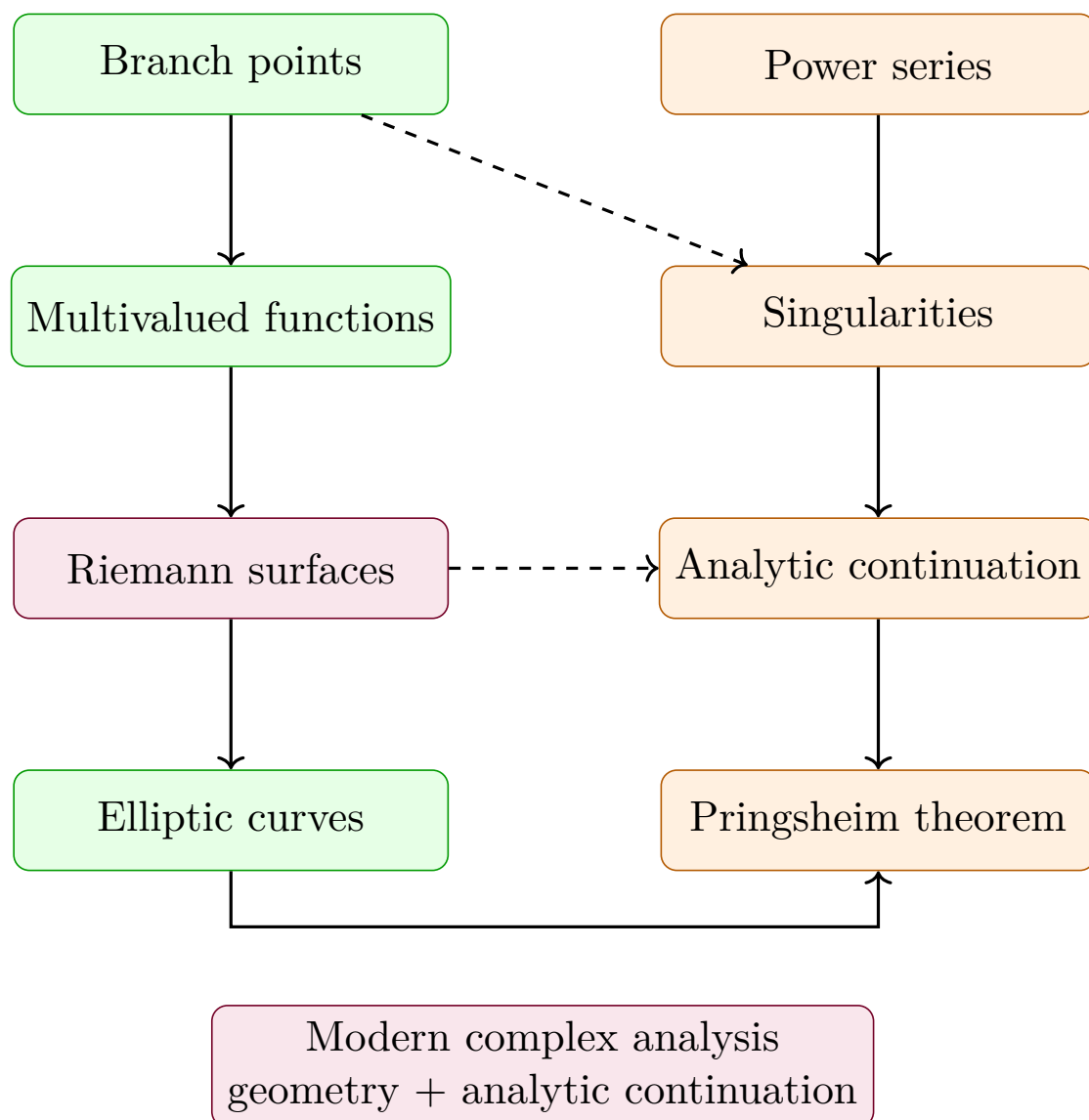
5.10 Positive and Non-Positive Coefficient Comparison

Series	Coefficients	Radius	Singularity on boundary
$\sum_{n=0}^{\infty} z^n$	$a_n = 1 \geq 0$	1	$z = 1$
$\sum_{n=0}^{\infty} 2^n z^n$	$a_n = 2^n \geq 0$	$\frac{1}{2}$	$z = \frac{1}{2}$
$\sum_{n=0}^{\infty} C_n z^n$	$C_n \geq 0$	$\frac{1}{4}$	$z = \frac{1}{4}$
$\sum_{n=0}^{\infty} (-1)^n z^n$	not nonnegative	1	$z = -1$

5.11 Problem 15 Versus Problem 16

	Problem 15	Problem 16
Main object	$w^2 = z^3 - z$	$\sum a_n z^n$
Main viewpoint	Geometry	Analysis
Key obstruction	Branch points	Singular boundary points
Main construction	Riemann surface	Taylor discs
Main theorem	Riemann–Hurwitz	Pringsheim theorem
Outcome	Elliptic curve	Positive real singularity
Future direction	Algebraic geometry	Analytic combinatorics

5.12 Final Concept Diagram



6 Examples Illustrating Pringsheim's Theorem

Recall Pringsheim's theorem.

Theorem 6.1 (Pringsheim). *Let*

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

be a power series with nonnegative coefficients

$$a_n \geq 0.$$

Suppose the radius of convergence is $R < \infty$. Then the point

$$z = R$$

is a singular point of f .

The theorem states that for power series with nonnegative coefficients, the first singularity necessarily lies on the positive real axis.

6.1 Example 1: The Geometric Series

Consider

$$f(z) = 1 + z + z^2 + z^3 + \cdots .$$

All coefficients satisfy

$$a_n = 1 \geq 0.$$

Step 1: Radius of Convergence

Using the Cauchy–Hadamard formula,

$$R = \frac{1}{\limsup_{n \rightarrow \infty} \sqrt[n]{1}} = 1.$$

Step 2: Sum of the Series

For $|z| < 1$,

$$f(z) = \frac{1}{1 - z}.$$

Step 3: Singularities

The denominator vanishes when

$$1 - z = 0,$$

hence

$$z = 1.$$

Therefore $z = 1$ is a pole.

Conclusion

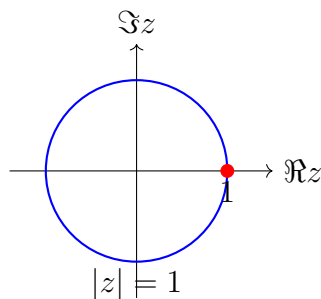
Since

$$R = 1,$$

the boundary point

$$z = R = 1$$

is singular, exactly as predicted by Pringsheim's theorem.



6.2 Example 2: The Logarithm

Consider

$$f(z) = \sum_{n=1}^{\infty} \frac{z^n}{n}.$$

All coefficients satisfy

$$a_n = \frac{1}{n} > 0.$$

Step 1: Radius of Convergence

Since

$$\lim_{n \rightarrow \infty} \sqrt[n]{\frac{1}{n}} = 1,$$

we obtain

$$R = 1.$$

Step 2: Closed Form

Inside the unit disk,

$$f(z) = -\log(1 - z).$$

Step 3: Singularity

The logarithm becomes singular at

$$1 - z = 0,$$

that is,

$$z = 1.$$

This is a branch point.

Conclusion

Again,

$$R = 1$$

and

$$z = R$$

is singular.



6.3 Example 3: An Entire Function

Consider

$$f(z) = \sum_{n=0}^{\infty} \frac{z^n}{(n!)^2}.$$

All coefficients are positive.

Radius of Convergence

Since

$$\sqrt[n]{\frac{1}{(n!)^2}} \rightarrow 0,$$

we obtain

$$R = \infty.$$

Interpretation

The function converges everywhere.

Indeed,

$$f(z) = I_0(2\sqrt{z}),$$

where I_0 is the modified Bessel function.

Conclusion

The function is entire.

Since $R = \infty$, Pringsheim's theorem does not apply because there is no finite boundary circle.

6.4 Example 4: Catalan Numbers

Consider the generating function

$$C(z) = \sum_{n=0}^{\infty} C_n z^n,$$

where

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

Step 1: Radius of Convergence

Using Stirling's formula,

$$C_n \sim \frac{4^n}{n^{3/2} \sqrt{\pi}}.$$

Therefore

$$R = \frac{1}{4}.$$

Step 2: Closed Formula

One has

$$C(z) = \frac{1 - \sqrt{1 - 4z}}{2z}.$$

Step 3: Singular Point

The square root becomes singular when

$$1 - 4z = 0.$$

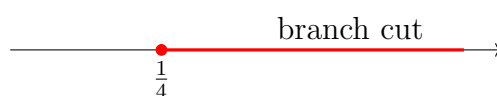
Hence

$$z = \frac{1}{4}.$$

Conclusion

The singularity occurs exactly at

$$z = R = \frac{1}{4}.$$



6.5 Example 5: Partition Numbers

Let

$$P(z) = \sum_{n=0}^{\infty} p(n)z^n,$$

where $p(n)$ denotes the number of integer partitions of n .

Euler discovered the product formula

$$P(z) = \prod_{m=1}^{\infty} \frac{1}{1 - z^m}.$$

All coefficients satisfy

$$p(n) \geq 0.$$

Radius of Convergence

One finds

$$R = 1.$$

Singularity

At

$$z = 1,$$

every factor

$$\frac{1}{1 - z^m}$$

becomes singular.

Therefore the product diverges.

Conclusion

The point

$$z = 1$$

is a singularity, exactly as predicted by Pringsheim's theorem.

6.6 Comparison Table

Function	Radius R	Singular at R ?	Type
$\frac{1}{1-z}$	1	Yes	Pole
$-\log(1-z)$	1	Yes	Branch point
$\sum \frac{z^n}{(n!)^2}$	∞	Not applicable	Entire
Catalan generating function	$\frac{1}{4}$	Yes	Square-root branch
Partition generating function	1	Yes	Infinite-product singularity

6.7 Summary

Pringsheim's theorem can be summarized as

Nonnegative coefficients $\implies$ a singularity occurs at the positive radius R .

This result is one of the central tools of analytic combinatorics and explains why the asymptotic behavior of counting sequences such as

$$C_n, \quad p(n), \quad \text{Motzkin numbers}, \quad \text{Fibonacci numbers},$$

is governed by the first singularity of their generating function on the positive real axis.

7 Pringsheim's Theorem and Modular Forms

Pringsheim's theorem is usually introduced in complex analysis and analytic combinatorics, but one of its deepest applications occurs in the theory of modular forms.

The bridge between the two subjects is provided by generating functions whose coefficients count arithmetic objects.

7.1 The Partition Function

Let

$$p(n)$$

denote the number of partitions of n .

For example,

$$4 = 4 = 3 + 1 = 2 + 2 = 2 + 1 + 1 = 1 + 1 + 1 + 1,$$

so

$$p(4) = 5.$$

Euler proved that the generating function is

$$P(q) = \sum_{n=0}^{\infty} p(n)q^n = \prod_{m=1}^{\infty} \frac{1}{1 - q^m}.$$

The coefficients satisfy

$$p(n) \geq 0.$$

Therefore Pringsheim's theorem applies.

Since the radius of convergence is

$$R = 1,$$

the point

$$q = 1$$

must be singular.

Indeed,

$$\prod_{m=1}^{\infty} \frac{1}{1 - q^m}$$

blows up as $q \rightarrow 1^-$.

Thus the dominant singularity occurs exactly at the positive real boundary point.

7.2 The Dedekind Eta Function

The infinite product above is closely related to the Dedekind eta function

$$\eta(\tau) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n),$$

where

$$q = e^{2\pi i\tau}, \quad \Im(\tau) > 0.$$

Hence

$$P(q) = \frac{q^{-1/24}}{\eta(\tau)}.$$

The partition generating function is therefore essentially the reciprocal of a modular form.

This remarkable observation connects

$$\text{Combinatorics} \quad \longleftrightarrow \quad \text{Modular Forms}.$$

7.3 Why the Singularity at $q = 1$ Matters

Pringsheim's theorem tells us where the dominant singularity is located:

$$q = 1.$$

Near this point,

$$P(q) = \prod_{m=1}^{\infty} \frac{1}{1 - q^m}$$

grows extremely rapidly.

The asymptotic growth of the coefficients

$$p(n)$$

is controlled by the behavior of the generating function near this singularity.

This principle is fundamental in analytic combinatorics:

Coefficient growth $\Longleftrightarrow$ Boundary singularities

7.4 Hardy–Ramanujan Asymptotics

Using the modular properties of the eta function, Hardy and Ramanujan proved

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right).$$

This is one of the most celebrated results in analytic number theory.

The argument begins with the singularity at

$$q = 1,$$

whose existence is guaranteed by Pringsheim's theorem.

Thus the logical chain is

Positive coefficients $\implies q = 1$ singular $\implies$ modular analysis $\implies$ asymptotics of $p(n)$.

7.5 Modular Forms Beyond Partitions

Many modular forms possess Fourier expansions

$$f(\tau) = \sum_{n=0}^{\infty} a_n q^n, \quad q = e^{2\pi i \tau}.$$

Examples include

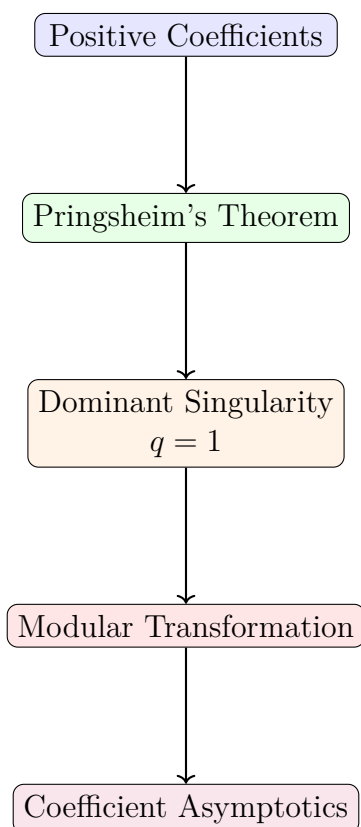
$$E_4(\tau), \quad E_6(\tau), \quad \Delta(\tau), \quad \eta(\tau).$$

Their coefficients encode deep arithmetic information:

Modular Form	Information Encoded
$\eta(\tau)$	Partitions
$E_4(\tau)$	Divisor sums
$E_6(\tau)$	Divisor sums
$\Delta(\tau)$	Ramanujan $\tau(n)$
Theta functions	Representations by quadratic forms

Although coefficients of modular forms are not always nonnegative, whenever a generating function has nonnegative coefficients, Pringsheim's theorem identifies the dominant boundary singularity and often provides the first step toward asymptotic analysis.

7.6 Conceptual Diagram



7.7 Big Picture

Pringsheim's theorem sits at the entrance to a vast circle of ideas:

Power Series $\rightarrow$ Singularities $\rightarrow$ Generating Functions $\rightarrow$ Modular Forms $\rightarrow$ Number Theory

The partition function is the classical example where a theorem from complex analysis, a generating function from combinatorics, and modular forms from number theory combine to produce one of the most beautiful asymptotic formulas in mathematics.